

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

**Time Duration: 1 Hour
Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I Steffi. J (19252448) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Steffi. J

94

94

RUBRICS FOR CALCULATION

Numerical Problems						Marks
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

20 + 20 + 23 + 18 + 13

794

bp

Employee Certification

Propositional Logic:

T = Kevin completed the training

C = Kevin receives certification

Rule $T \rightarrow C$

Convert into CNF: $T \rightarrow C$ $\neg T \vee C$

Negate Logic: $\neg C$

Resolution: $(\neg T \vee C)$

Final solution: Since the empty clause is obtained
Kevin receives certification.

2) Medical Attention Alert

Propositional Logic:

F = Ravi has a fever

M = Ravi needs medical attention

$F \rightarrow M$

CNF conversion: $\neg F \vee M$

Negate the Goal: $\neg M$

Resolution: $\neg F \vee M$

Final conclusion: Ravi needs medical attention

3) Library Late Fine

Propositional Logic

L = Book was returned late

F = Fine is charged

Rule $L \rightarrow F$

CNF conversion: $L \rightarrow F$

$\neg L \vee F$

Negate the Goal : $\neg F$

Resolution : $\neg L \vee F$

Final conclusion : A fine is charged

4) Flight Delay Notification

Propositional Logic :

D = Flight 2020 is delayed

N = Passengers are notified

Rule $\rightarrow D \rightarrow N$

convert into CNF : $\neg D \vee N$

Negate the Goal : $\neg N$

Resolution : $\neg D \vee N$

Final conclusion : Passengers are notified.

5) Vehicle Registration process

Propositional Logic :

E = Car passed the emissions test.

R = ~~car~~ is certified roadworthy

G = ~~car~~ can be registered

Rule $E \rightarrow R$, $R \rightarrow G$

Fact : E

Goal : G

Convert implications Into CNF

$$E \rightarrow R$$

Remove implication : $\neg E \vee R$

Rule 2 : $R \rightarrow G$

Remove implication : $\neg R \vee G$

Fact : E

CNF clauses are :

1) $\neg E \vee R$

2) $\neg R \vee G$

3) E

Negate the Goal : $\neg G$

Final set of clauses

$$\neg E \vee R$$

$$\neg R \vee G$$

$$E$$

$$\neg G$$

Resolution

Step 1 $\neg E \vee R$

Resolve E and $\neg E \dots R$

Therefore : The car is certified Roadworthy

Step 2 $\neg R \vee G$

Resolve R and $\neg R \dots G$

Therefore : The car can be registered.